

Business Insights Report

Data Analytics & Business Intelligence — Task 1

Intern: Nidhi Sharma | Maincrafts Technology | June 2026

Dataset: [Sample Superstore](#) | Tools: Microsoft Excel, SQLite, Power BI

1. Regional Sales Performance

The **West region leads all regions** with \$725,457 in total sales, contributing 31.58% of overall revenue. East follows at \$678,781 (29.55%), Central at \$501,239 (21.82%), and South at \$391,721 (17.05%). The performance gap between the highest and lowest performing regions is nearly 2x, indicating significant regional disparity requiring strategic attention. These findings were independently validated through SQL query analysis.

2. Category Profitability Analysis

Technology is the most profitable category, generating \$145,454 in profit and representing 50.8% of total business profit. Office Supplies contribute \$122,490 (42.8%). Furniture is critically underperforming at just \$18,451 — only 6.4% of total profit — despite accounting for a significant share of sales volume. Average profit margin analysis further confirms this: Technology at 15.61%, Office Supplies at 13.80%, and Furniture at a dangerously low 3.88%.

3. Discount Impact on Profitability

Discount analysis reveals a **direct and severe negative relationship between discount levels and profitability**. SQL-validated results show:

Discount Level	Average Profit per Transaction
0%	+\$66.90
20%	+\$24.70
30%	-\$45.68
50%	-\$310.70
70%	-\$95.87
80%	-\$101.80

The 20% discount level represents the **profitability breakeven point** — every transaction discounted beyond this threshold generates a net loss. Overall, **18.72% of all transactions — nearly 1 in 5 sales — resulted in negative profit margins**, representing a systemic pricing risk embedded within otherwise healthy revenue figures.

4. Monthly & Seasonal Sales Trends

November is the strongest-performing month at \$352,461, followed closely by December at \$325,293 and September at \$307,649. February records the lowest monthly sales at \$59,751—approximately 6x lower than the peak month. Q4 (October–December) consistently outperforms all other quarters across all years in the dataset, confirming a structural seasonal demand pattern rather than a one-time occurrence. This predictable pattern presents a clear opportunity for proactive inventory and marketing planning.

5. Strategic Recommendations for Management

5.1 Implement a Maximum Discount Policy

Data proves that discounts above 20% consistently produce losses. A maximum 20% discount cap should be implemented immediately across all categories, with documented management approval required for any exceptions.

5.2 Address South Region Underperformance

South region contributes only 17.05% of revenue. A root cause analysis covering distribution gaps, pricing alignment, and brand awareness should be conducted, followed by a targeted regional marketing initiative.

5.3 Urgent Furniture Category Review

Furniture's 3.88% average profit margin signals critical margin erosion. A full discount audit of Furniture transactions is recommended alongside a pricing restructure and removal of consistently loss-making SKUs.

5.4 Maximize Q4 Revenue Opportunity

With November consistently peaking, all inventory procurement, staffing decisions, and marketing budgets should be finalized and allocated by September to capitalize fully on Q4 demand.

5.5 Protect Technology Category Performance

Technology drives 50.8% of total profit at a 15.61% margin. This category should be protected from aggressive discounting and prioritized for product line expansion.

5.6 Introduce a Key Account Program

Top customer Sean Miller generates \$25,043 individually. The top 10 customers collectively contribute \$153,811 — 6.7% of total revenue. A structured key account management program with loyalty incentives would protect and grow this high-value segment.
